PosturePal

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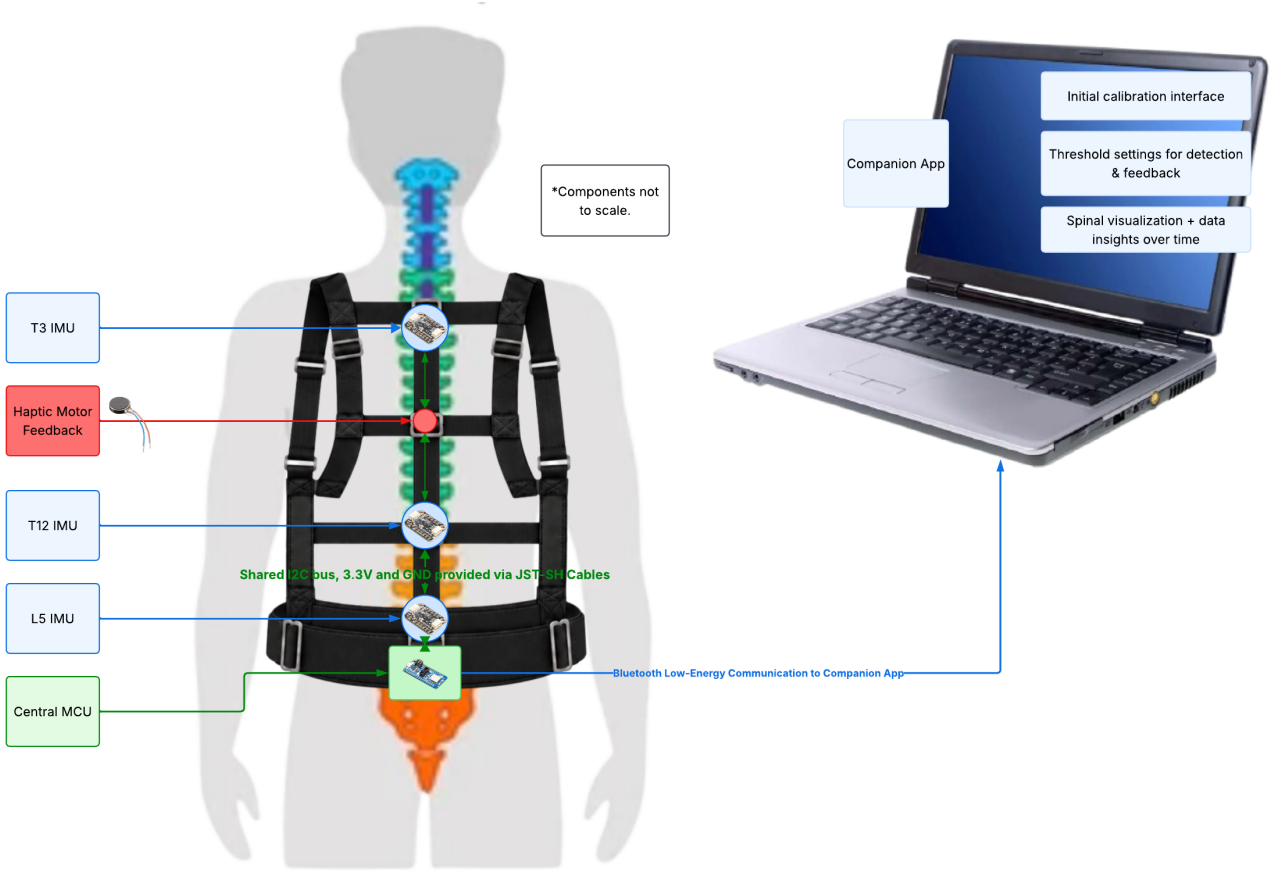


Figure 1: PosturePal system overview, showing sensor placement, haptic feedback, and the companion application.

Abstract— PosturePal is a wearable system that monitors seated posture and provides vibration reminders to encourage users to adjust their posture during desk work. Three motion sensors mounted along the back measure upper- and lower-back bending relative to a personalized baseline. When a deviation persists beyond the user's chosen limits, the system provides feedback without physically restricting movement. A companion desktop application guides calibration, allows users to customize reminders, and visualizes posture changes throughout their work session.

Index Terms— Haptic feedback, inertial measurement unit, posture monitoring, sensor fusion, wearable electronics.

1 INTRODUCTION

Students and office workers who spend long periods at computers often find it difficult to notice and correct sustained changes in their seated posture. PosturePal aims to provide personalized reminders that support posture awareness without physically restricting movement.

Existing approaches include mechanical braces and electronic feedback devices. ComfyBrace uses adjustable shoulder straps for mechanical support [1], which may restrict movement rather than simply remind users to adjust their posture. UPRIGHT GO 2 provides vibration reminders and application-based tracking, but measures posture at a single upper-back location and uses adhesive mounting that requires periodic replacement [2]. These limitations motivate a reusable wearable that measures bending across multiple back regions while allowing users to move freely.

PosturePal combines a vest containing three inertial measurement units (IMUs), a microcontroller, battery, and haptic actuator with a desktop application. Sensors at the upper thoracic, lower thoracic, and lower lumbar regions provide estimates of upper- and lower-back bending. Users calibrate a comfortable neutral posture, then choose how far and how long their posture may deviate before receiving

a vibration reminder.

The desktop application guides calibration, supports threshold adjustments, and displays live posture and measurement history. By comparing measurements with a personalized reference, PosturePal helps users recognize sustained posture changes without claiming to determine whether a posture is medically correct.

2 USE CASE REQUIREMENTS

2.1 Posture Measurements

PosturePal must accurately measure lower and upper back orientation during calibration, threshold setting, and continuous monitoring. Both regional-angle estimates must have a root mean square error (RMSE) of less than 5° against independently measured reference angles under the test conditions in Section 7. These angles describe bending between sensing locations, rather than the exact curvature of individual vertebrae. Paloschi et al. validated three magneto-inertial sensors placed at T3, T12, and S1 against optical motion capture during forward bending and sit-to-stand movements, reporting regional-angle RMSE values of 3.0–5.6° [3]. Hence, to meet results of existing research, we aim for a RMSE of less than 5° in regional-angle estimations.

2.2 Haptic Feedback

After calibration, PosturePal must detect at least 90% of intended alert trials in which a regional deviation exceeds its configured angle threshold continuously for the configured duration. The system must identify the region and activate its single haptic motor within 1 second after that duration ends. At least 98% of delivered alerts must be felt by the user through typical clothing which will be verified through a user survey.

2.3 Visual Feedback

The delay between a physical posture change and the corresponding display update must be less than 250 ms to keep visual feedback responsive. Nielsen's response-time guidance places immediate responses near 100ms and uninterrupted interaction within 1s [4]. A separate study found that visual delays of 250ms and 500ms affected dynamic standing responses [5], suggesting that delays in this range can influence how users respond to posture feedback. Although that study examined standing in virtual reality rather than seated posture, it supports minimizing feedback delay. We therefore selected 250ms as an engineering target for responsive monitoring, rather than a clinically established limit, and will verify it through end-to-end system testing.

2.4 User Comfort, Accessibility, and Safety

To encourage prolonged use, PosturePal must weigh less than five pounds, and maintain a surface temperature below 34°C, with no hotspots exceeding this limit. Temperature will be evaluated at the user-facing electronics, battery pocket, and motor surfaces in an indoor ambient temperature of 22°C ± 2°C, as described in Section 7. These requirements are intended to minimize physical discomfort when wearing the device throughout the day. Moreover, to accommodate a broad range of body sizes, the vest must be adjustable for users of sizes small to extra-large.

Finally, PosturePal primarily targets office workers and students who spend extended periods working at computers, often for eight hours or more per day. Consequently, the device must provide at least eight hours of continuous battery life to support a full workday without requiring recharging. Collectively, these requirements are intended to encourage consistent daily use, as discomfort, frequent adjustments, or interruptions for charging could discourage users from wearing the device and undermine its long-term effectiveness.

2.5 Broader Societal Considerations

Public health and safety influenced the decision to design PosturePal as a feedback device rather than a mechanical brace. The vest encourages the user to actively adjust posture without physically forcing the spine into a particular configuration. User-defined calibration also avoids presenting a single universal spinal configuration as medically correct. The intended use case is for a clinician to assist in defining a user's neutral seated posture. Also, accessibility influenced the adjustable small–extra large vest requirement and the use of haptic rather than exclusively visual feedback. The adjustable design should also reduce the need to manufacture separate electronics for every body size, potentially reducing cost and material use.

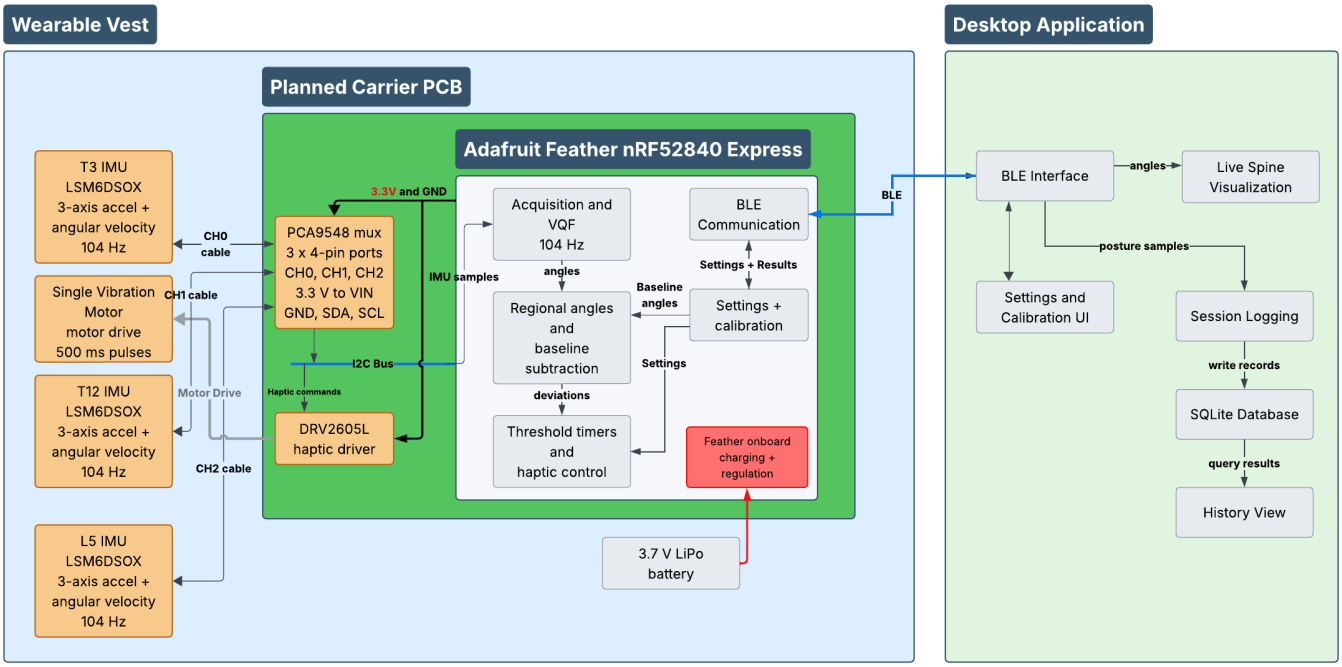


Figure 2: PosturePal system architecture, showing the wearable hardware, firmware processing, and desktop application.

3 ARCHITECTURE AND/OR PRINCIPLE OF OPERATION

3.1 System Overview

Figure 2 summarizes the wearable layout and companion application. PosturePal measures posture by comparing the orientations of three sensing locations along the spine, near T3, T12, and L5. These locations provide information about the upper and lower portions of the back, and comparing the orientations at neighboring sensing locations allows PosturePal to detect changes in upper- and lower-back bending.

The relative angles between sensing locations approximate changes in upper- and lower-back shape without reconstructing the exact curvature of individual vertebrae. PosturePal compares these angles with reference values recorded during calibration in the user's preferred posture. If a deviation exceeds the user-selected threshold for a specified duration, the system provides a haptic reminder while continuously sending posture measurements to the desktop app. This approach provides personalized feedback on posture changes without classifying whether the user's posture is medically correct.

3.2 Orientation Estimation

PosturePal estimates regional back bending by comparing the orientations of neighboring sensing locations, as illustrated in Figure 3. Each IMU location supplies motion measurements from which the system estimates its orientation relative to gravity. In general, gyroscopes capture short-term changes in rotation but accumulate integration drift, whereas accelerometers provide a gravity reference but can be distorted by movement. Combining these complementary signals allows more stable tilt estimates.

Orientation estimates are expressed as rotations in three-dimensional space. Because slouching primarily involves forward and backward bending, PosturePal evaluates these tilt angles in the sagittal plane. Let θ_{T3}, θ_{T12}, and θ_{L5} denote the sagittal-plane tilt angles at the three locations. The differences between adjacent tilt angles define the upper- and lower-back relative angles:

$$\theta_{\text{upper}} = \theta_{T3} - \theta_{T12}, \tag{1}$$

$$\theta_{\text{lower}} = \theta_{T12} - \theta_{L5}. \tag{2}$$

The upper and lower measurements represent the differences between T3 and T12 and between T12 and L5, respectively. The selected sensor hardware, detailed VQF pipeline, and angle-extraction calculations are described in Section 6.

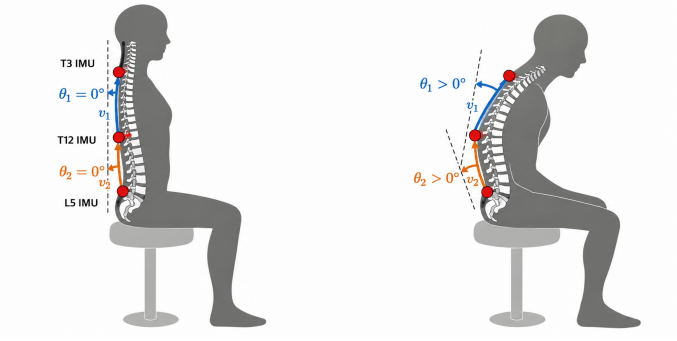


Figure 3: Illustration of neutral seated posture and forward bending at the three sensing locations.

3.3 Calibration: Establishing a Personalized Reference

Because natural spinal curvature and comfortable seated posture vary between individuals, PosturePal uses a personalized calibration process rather than assuming that a particular spinal angle represents poor posture for every user. Posture quality is defined operationally as the magnitude and duration of changes in regional bending relative to the user's calibrated reference. This provides a personalized measure of posture deviation rather than an absolute medical assessment of spinal alignment.

During initial calibration, the desktop application instructs the user to assume and maintain their preferred neutral seated posture. While the user holds this position, the three IMUs continuously measure their orientations, allowing the system to estimate the relative angles of the upper and lower back. Once these measurements have stabilized, PosturePal records the upper- and lower-back angles as the user's baseline values. These values represent the user's preferred posture and establish a personalized zero-deviation reference for each region.

At runtime, the firmware computes the signed deviations:

$$\Delta\theta_{\text{upper}} = \theta_{\text{upper,current}} - \theta_{\text{upper,base}}, \quad (3)$$

$$\Delta\theta_{\text{lower}} = \theta_{\text{lower,current}} - \theta_{\text{lower,base}}. \quad (4)$$

The calibrated posture therefore produces zero *deviation*, even when the regional angles themselves are nonzero. For example, an upper-back reference angle of 12° and a current angle of 19° produce a deviation of +7°. Calibration establishes a reference for comparison without changing the measured angles. The same calculation is performed independently for the lower back.

3.4 Threshold Configuration

After establishing their baseline posture, users configure separate angular thresholds for the upper and lower back, along with the duration for which a deviation must persist before triggering feedback. The application's 3D spine visualization provides immediate visual feedback, helping users understand how their movements affect the measured regional angles. Users can subsequently adjust the thresholds and feedback duration to accommodate differences in anatomy, everyday movement, and feedback preferences. The application also provides a movement-check procedure to help users identify whether common movements, such as turning or reclining, would unintentionally trigger reminders. These adjustments can be made without changing the original zero-deviation reference unless the user chooses to recalibrate. The detailed interface-guided calibration and movement-check procedures are described in Section 6.

3.5 Posture Detection and Haptic Feedback

The microcontroller continuously evaluates posture against the calibrated reference and configured settings. A reminder is generated only when the magnitude of a regional deviation exceeds its configured angular threshold continuously for the required duration. Brief movements reset the timer when the angle returns within the threshold.

When a deviation exceeds the selected threshold for the specified duration, the system activates a haptic reminder to prompt the user to adjust their posture. Posture detection and haptic feedback operate independently of the desktop application, allowing reminders to continue when the interface is not open.

3.6 Desktop Application and Visualization

The desktop application allows users to calibrate their preferred posture, configure the angular deviation thresholds and reminder duration, view live posture measurements, and review posture history. During normal use, an approximate spine visualization shows changes in upper- and lower-back bending relative to the calibrated reference, helping the user understand which region has deviated.

The desktop application receives posture measurements for visualization and communicates updated user settings to the microcontroller. Details of the wireless protocol, message formats, data storage, and interface components are provided in Section 6.

4 DESIGN REQUIREMENTS

The design requirements translate the use-case requirements in Section 2 into quantitative targets for the sensing, feedback, desktop, and wearable subsystems. Table 1 summarizes these targets and explains how their integration supports accurate posture monitoring, timely reminders, responsive visualization, and comfortable full-day use. Integrated performance will be evaluated using the tests in Section 7.

Table 1: Key design specifications and related use-case requirements.

Subsystem	Key design specifications	Use-case requirement supported
Posture measurement	Three IMUs at 104 Hz; 5 s averaged calibration; mount rotation $\leq 2^\circ$, shift ≤ 5 mm.	Accurate upper/lower-back measurements: RMSE $< 5^\circ$.
Haptic feedback	Regional threshold timers; nonblocking 500 ms pulses; motor onset ≤ 1 s after the configured duration.	Detect $\geq 90\%$ of intended alerts; $\geq 98\%$ felt; $\geq 95\%$ of ordinary-movement trials without unwanted alerts.
Desktop app	25 updates/s; 210 ms total sensing-to-display delay budget.	Responsive visual feedback: display delay $<$ 250 ms.
Vest and fit	Adjustable S–XL mounts; working mass target < 5 lb; contact surfaces $< 34^\circ$C.	Comfortable, accessible wear: S–XL fit, mass < 5 lb, and surfaces $< 34^\circ$C.
Power	Proposed 2500 mAh battery; average current ≤ 200 mA.	Full-day use: battery life ≥ 8 h.

4.1 Posture Measurement Accuracy

Use-case link: Measure upper- and lower-back regional angles with RMSE below 5° during calibration and continuous monitoring.

The mounts must hold the three sensing nodes near T3, T12, and L5 and prevent them from shifting or rotating relative to the garment. To evaluate overall measurement accuracy, PosturePal's regional-angle measurements must be compared with an independent reference measurement, accounting for errors introduced by sensing, mounting, filtering, and calibration.

$$\mathrm{RMSE}_r = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(\widehat{\theta}_{r,i} - \theta_{r,i}^{\mathrm{ref}}\right)^2} < 5^\circ, \tag{5}$$

where $r \in \{\text{upper}, \text{lower}\}$ and N is the number of paired system/reference samples.

4.2 Posture Classification and Haptic Feedback

Use-case link: Provide timely, perceptible reminders for sustained deviations while minimizing unwanted alerts during ordinary seated activity.

The MCU monitors how long each back region remains beyond its configured angular threshold. If either region exceeds its threshold for the selected duration, the motor begins providing repeated 500 ms vibration pulses, separated by a brief pause, until both regions return within their thresholds. Returning within a region's threshold resets its duration timer, so a later deviation must persist for the selected duration before triggering another reminder. Posture sensing continues throughout the feedback period.

Pausing monitoring, starting recalibration, or receiving invalid sensor data stops the vibration and resets the duration timers. Invalid sensor data also causes the desktop application to indicate that live posture measurements are unavailable. If the BLE connection is lost, the wearable continues local monitoring and feedback using the most recently confirmed settings.

Once a posture deviation has remained beyond its configured threshold for the selected duration, haptic feedback must begin within 1s. This response time includes acquiring and transferring sensor measurements, evaluating the deviation, and activating the motor. The equation for this is as follows:

$$t_{\mathrm{sample}} + t_{\mathrm{transfer}} + t_{\mathrm{classify}} + t_{\mathrm{motor}} \leq 1 \text{ s}. \tag{6}$$

The system must correctly trigger feedback in at least 90% of trials designed to produce an alert, and users must report feeling at least 98% of delivered vibration pulses. During ordinary-movement testing, at least 95% of trials must produce no unwanted alerts.

4.3 Desktop App

Use-case link: Show posture changes within 250ms so users can interpret their movements and configure personalized feedback.

The desktop application must display posture changes within 250ms so users can interpret their movements and configure personalized feedback. This limit includes acquisition, filtering and MCU processing, BLE delivery, and application rendering:

$$t_{\mathrm{sample}} + t_{\mathrm{MCU}} + t_{\mathrm{BLE}} + t_{\mathrm{render}} < 250 \text{ ms}. \tag{7}$$

The display will target 25 posture updates per second while the MCU samples at the higher IMU rate. Configuration changes will be confirmed only after MCU acknowledgment, and the interface will clearly indicate disconnected or stale data.

4.4 Vest Implementation

Use-case link: Support comfortable, adjustable S–XL wear and at least eight hours of continuous operation.

The harness must allow independent vertical adjustment of all three sensing nodes across the tested S–XL

sizes. Each pocket will include a repeatable orientation mark and strain relief so cable tension does not rotate the board. The combined vest, battery, wiring, and electronics must remain below the 5 lb maximum.

All parts must support an eight-hour session with continuous sensing, BLE, and the stated motor duty cycle. Section 6.2 gives the estimated current budget and the measurement needed to confirm battery capacity.

5 DESIGN TRADE STUDIES

5.1 Fixed "Correct" Posture vs. User-Defined Calibration

Another possible approach was to define a fixed set of angles corresponding to "good" and "bad" posture. This would simplify the classification algorithm, but would assume that the same spinal configuration is appropriate for all users. PosturePal instead performs user-specific calibration and detects deviation from the user's neutral baseline. This makes the system adaptable to differences in anatomy, sitting style, and sensor placement. It also limits the system's medical claims: the device does not determine whether a user's baseline posture is clinically ideal, but rather detects when posture changes from the baseline selected by that user.

The tradeoff is that the system depends on a reliable calibration procedure. Incorrect neutral-position calibration shifts the reference used for all subsequent comparisons and can create false feedback. The calibration procedure must therefore establish a stable, repeatable baseline.

5.2 Posture Detection: Relative-Angle Thresholds vs. Learned Classification

Two approaches were considered for posture detection: a relative-angle thresholds using multiple sensors, and a trained classifier. A trained classifier could potentially identify more complex movement patterns, but would require representative labeled IMU data, training and validation across users, and additional embedded processing and maintenance.

We selected relative-angle thresholds for the prototype because their behavior is easier for users to calibrate and for us to verify. The method subtracts adjacent sagittal tilt estimates and compares the resulting regional angles with the user's calibrated reference values. This provides two interpretable posture measures from three sensors and supports quantitative verification against known angles.

The tradeoff is that relative-angle thresholds cannot always distinguish sustained undesirable posture from intentional movements that produce similar bending. The adjustable duration threshold and guided movement check partially address this limitation. Testing will quantify false reminders during ordinary actions and determine whether the rule-based approach satisfies the stated requirements.

5.3 I^2C vs. SPI

Both I^2C and SPI were considered for communication with the distributed IMUs. SPI provides higher bandwidth and allows each identical sensor to be addressed using a separate chip-select signal. However, this would require additional conductors to each sensor location, increasing wiring complexity through the wearable. I^2C requires only shared data and clock lines and therefore simplifies routing to multiple spatially separated sensors. Its lower bandwidth is still expected to be sufficient for the relatively small IMU data stream. The disadvantage is the limited number of selectable LSM6DSOX addresses, requiring the additional PCA9548 multiplexer. Because wearable wiring complexity was considered more important than maximizing bus bandwidth, I^2C with a multiplexer was selected. The final I^2C bus will be configured to operate at 400 kHz. Based on the expected transaction size for reading the six accelerometer and gyroscope channels from each LSM6DSOX, the estimated bus-read delay between the first and third IMU is approximately 0.5–1.0ms. Testing will verify the delay and evaluate its effect on regional-angle accuracy against the 5° RMSE requirement.

5.4 MCU and Power Integration

Two approaches were considered for the final electronics design: integrating a bare nRF52840 directly onto the custom PCB or retaining the Adafruit Feather nRF52840 Express as a module on a carrier PCB. A fully integrated nRF52840 design would reduce board area and allow greater control over the power, charging, RF, and connector layout. However, it would also require significantly more PCB design effort, including implementation of the MCU support circuitry, battery charging, regulation, programming/debug interfaces, and BLE antenna layout. Because of the limited development and fabrication schedule, the Feather-based design was selected. Although it increases board size and cost, it reuses already validated charging, regulation, USB, BLE, and MCU circuitry, reducing routing complexity and bring-up risk. The custom PCB can therefore focus on integrating the PCA9548 multiplexer, haptic driver, power distribution, and IMU connectors.

5.5 Haptic Motor Placement

We initially considered placing separate haptic motors near each sensing region so that feedback could be localized to the upper thoracic, lower thoracic, and lumbar areas. However, this approach would require three haptic drivers, additional bus devices, and significantly more wiring throughout the garment. In particular, a motor next to the T3 sensor would require a relatively long cable from the main electronics near the waist or lower back. Furthermore, vibrations from the haptic motors could disturb nearby IMU measurements, particularly if the motors were placed close to the sensors, introducing additional noise into the posture estimates. The design therefore uses a sin-

gle haptic motor located between the T3 and T12 IMUs. It will be connected to the DRV2605L driver using a flexible two-wire cable with a detachable connector and appropriate strain relief. The tradeoff is that feedback is no longer spatially localized, as now a single vibration alert is used to notify the user that their posture has exceeded the configured threshold.

6 SYSTEM IMPLEMENTATION

6.1 Hardware Implementation

Figure 4 shows the planned sensing hardware and connections. PosturePal's planned hardware uses three LSM6DSOX six-degree-of-freedom IMUs, a Nordic nRF52840 microcontroller, a PCA9548 I^2C multiplexer, a DRV2605L haptic motor driver, and a disc vibration motor. Breakout boards are being used during early development so that individual subsystems can be validated independently before the final custom PCB is assembled. Breakout modules can be retained during integration if fabrication or bring-up is delayed.

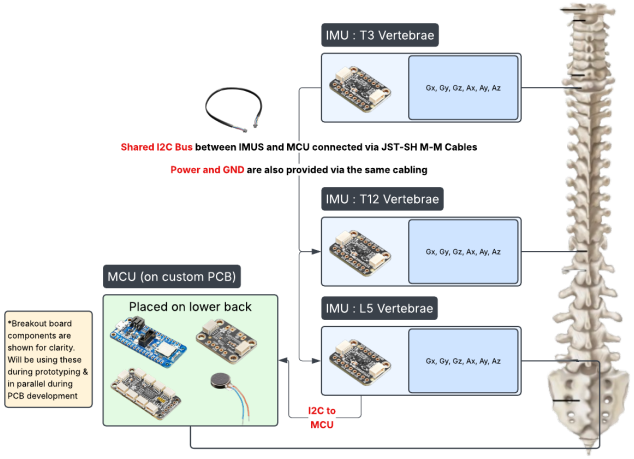


Figure 4: Planned hardware layout and connections between the three IMUs and central electronics. The hardware illustrations simplify cable routing for visual clarity. The illustrated routing does not represent the final electrical connections.

IMUs. In the final wearable, three LSM6DSOX IMUs will be mounted along the vest near T3, T12, and L5. Each sensor generates three acceleration measurements (a_x, a_y, a_z) and three angular-velocity measurements $(\omega_x, \omega_y, \omega_z)$. These values are read digitally by the central microcontroller and supplied to the VQF implementation.

I^2C Interface and Multiplexer. The IMUs communicate with the MCU using I^2C. Because the LSM6DSOX supports only two selectable addresses, 0x6A and 0x6B, a PCA9548 multiplexer places the three sensors on separate channels. The MCU selects each channel in turn to read its sensor.

Sequential reads introduce a small acquisition-time offset between sensors, which may affect relative-angle accuracy during movement. The MCU will timestamp the measurements, and testing will measure the maximum inter-sensor acquisition delay and evaluate its effect on posture measurements.

MCU. The Adafruit Feather nRF52840 Express serves as the central embedded processor. Its firmware will read and timestamp measurements from the three IMUs, run VQF orientation estimation, calculate regional angles, compare deviations with calibrated thresholds, control haptic feedback, and manage BLE communication with the desktop application.

The firmware is being developed in Arduino/C++, using the Adafruit LSM6DS and DRV2605 libraries for sensor communication and motor control, respectively, and the C++ implementation of VQF for orientation estimation. All source code will be maintained in a shared GitHub repository for version control and team collaboration.

Haptic Hardware. Haptic feedback is produced using a DRV2605L motor driver and disc vibration motor. The MCU communicates with the DRV2605L over I^2C and commands vibration when one of the posture thresholds is exceeded. A dedicated haptic driver is used instead of driving the motor directly from a microcontroller GPIO because the motor requires higher current than an MCU output pin can safely provide and benefits from dedicated waveform-control circuitry. The haptic motor will be placed between the T3 and T12 IMUs. See Section 5.5 for more information on how this placement was determined.

6.2 Power Implementation

PosturePal will use a rechargeable 3.7 V LiPo connected to the Feather's battery input, using its onboard charging and voltage-regulation circuitry. A 2500 mAh battery is proposed, although the final capacity will be selected based on measured current consumption and testing.

Table 2: Preliminary average-current budget. Values are planning allocations informed by component specifications and an assumed motor duty cycle.

Load	Allocation (mA)
Feather MCU and BLE	50
Three IMU modules	15
Mux module	2
Haptic driver, motor off	3
LEDs, losses, and other board loads	10
Motor: 80 mA, 0.5 s every 30 s	1.33
Estimated average	81.33

These allocations are preliminary estimates of battery-input current. Reserving 30% of rated capacity gives an estimated runtime of

$$t_{\text{est}} = \frac{0.70(2500\ \text{mAh})}{81.33\ \text{mA}} \approx 21.5\ \text{h}. \tag{8}$$

At an average draw of 200 mA, the estimated runtime becomes 8.75 hours.

6.3 Firmware Implementation

The planned firmware converts motion measurements into posture estimates, compares them with the user's calibrated posture, and decides when to provide a reminder. These calculations run locally on the MCU.

1. Acquire sensor readings. Each of the three IMUs measures acceleration (a_x, a_y, a_z) and angular velocity $(\omega_x, \omega_y, \omega_z)$ at a sampling frequency of 104 Hz. Acceleration provides information about gravity and movement and angular velocity describes how quickly the sensor rotates. The MCU reads these measurements over I^2C.

2. Combine readings using VQF. The MCU runs a separate VQF filter for each IMU. VQF combines gyroscope readings that track rotation with accelerometer readings that provide a gravity reference. It also estimates gyroscope bias to reduce drift, allowing tilt estimation without a magnetometer.

VQF outputs a quaternion, $q = (w, x, y, z)$, which uses four numbers to describe the sensor's 3D orientation. Together, these components describe how far the sensor is rotated and the axis of that rotation.

Intuitively, the rotation axis is an imaginary axle around which the sensor turns. The component w is related to the amount of rotation. The components x, y, and z describe how that axle points along the three coordinate directions, scaled according to the rotation amount. For example, a rotation purely about the y axis has zero x and z components, while w and y describe the rotation.

We retain this representation to track combined movements, such as bending while turning, before extracting the forward/backward tilt needed for posture monitoring.

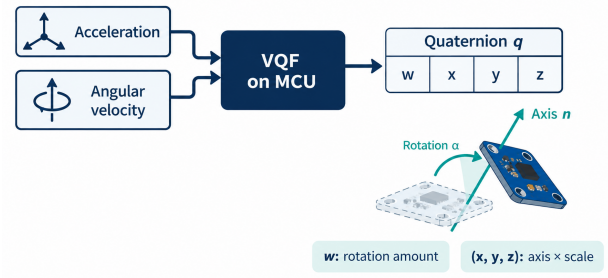


Figure 5: VQF produces a quaternion describing rotation about an axis. Generated using ChatGPT [10].

3. Convert orientation into back tilt. The quaternion already represents orientation. The firmware converts it into a rotation matrix R_i so it can express the upward direction in each sensor's body axes:

$$\mathbf{u}_i = R_i^T \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \qquad \theta_i = \operatorname{atan2}(-u_{i,x}, u_{i,z}).$$

Here, R_i maps body axes into the gravity-aligned world frame. The calculation uses the forward and upward components of $\mathbf{u}_i$ to find forward/backward tilt (or just the sagittal plane).

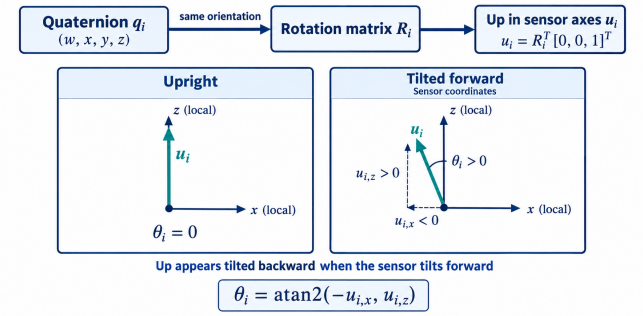


Figure 6: From quaternion orientation to tilt: world up expressed in sensor coordinates for upright and forward-tilted sensors. Generated using ChatGPT [10].

4. Compare with the baseline and decide. Subtracting neighboring tilts gives $\theta_{\text{upper}} = \theta_{T3} - \theta_{T12}$ and $\theta_{\text{lower}} = \theta_{T12} - \theta_{L5}$. For each region, firmware then calculates $\Delta\theta = \theta_{\text{current}} - \theta_{\text{baseline}}$.

If the magnitude of either deviation exceeds its configured threshold continuously for the selected duration, the MCU activates haptic feedback. Returning within threshold resets that region's timer. Posture updates are also sent over BLE.

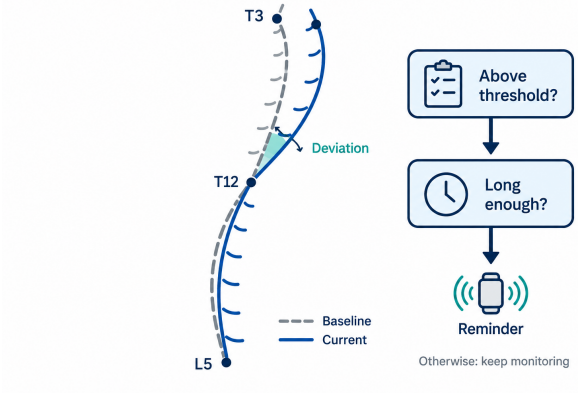


Figure 7: Illustrative baseline/current back positions and reminder decision. Generated using ChatGPT [10].

6.4 BLE Communication

BLE connects the vest to the desktop application, carrying user settings and commands to the MCU and returning posture measurements and confirmation messages. Table 3 summarizes the planned messages (direction is relative to the desktop app).

Table 3: Planned BLE messages in typical workflow order.

Message	Direction	When sent
`calibrate`	Outgoing	When the user starts calibration.
`calibration_res`	Incoming	When calibration finishes, returning reference angles or failure status.
`set_thresholds`	Outgoing	When the user sets or adjusts angular thresholds and reminder duration.
`settings_ack`	Incoming	After the MCU applies the updated settings.
`posture_update`	Incoming	At 25 Hz during monitoring, carrying regional angles and deviations.
`pause_resume`	Outgoing	When the user pauses or resumes monitoring.
`monitoring_ack`	Incoming	After the MCU applies the requested monitoring state.

The small message payloads are expected to fit within BLE's available bandwidth. The system will target a connection interval of 15–30 ms, subject to the desktop device's supported parameters. The complete sensing-to-display pipeline must update the visualization within 250 ms of a physical posture change.

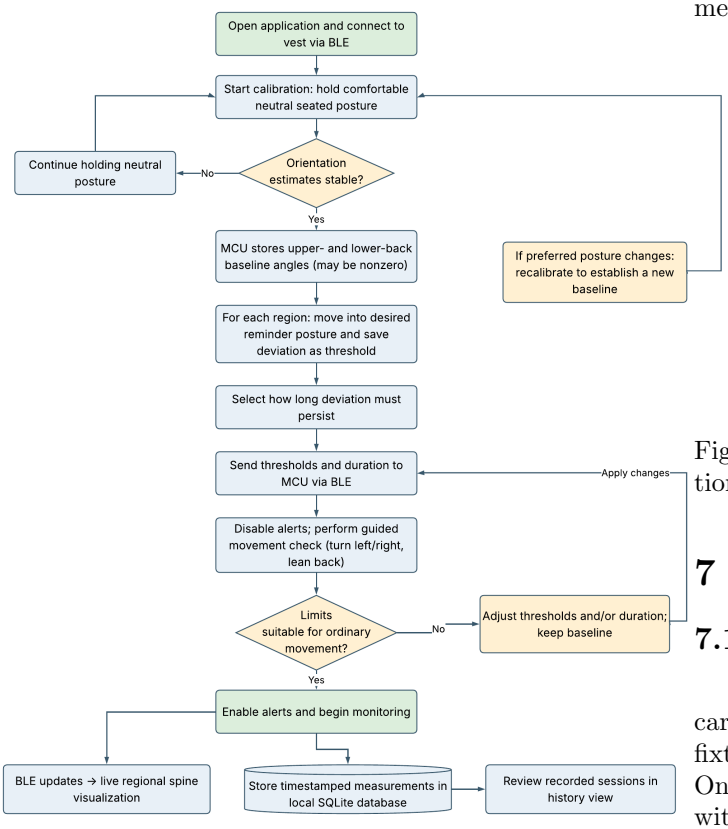


Figure 8: Initial and later software workflow.

6.5 Desktop Software

The application will include a settings manager, BLE interface, live posture display, history view, and local SQLite database. During calibration, it asks the user to maintain a comfortable neutral seated posture while the orientation estimates stabilize. The MCU then stores the upper- and lower-back relative angles as baseline values, which may be nonzero.

With alerts disabled, a guided movement check shows whether ordinary motions, such as turning left or right and leaning back, cross the configured limits. Users can adjust thresholds and duration without recalibrating, or establish a new baseline if their preferred posture changes.

BLE updates drive the live spine visualization, while timestamped measurements are stored in SQLite for historical analysis. The visualization approximates regional back shape from the three measured orientations.

6.6 Mechanical/Wearable Implementation

The electronic sensing nodes will be integrated into an adjustable vest/harness. The vest will be modified so that the IMUs remain securely positioned over the intended T3, T12, and L5 locations while accommodating users across sizes small–extra large. Adjustable mounting straps or pockets will be used so that the IMU position can be changed to account for differences in body size. Figure 9 illustrates the proposed harness layout and sensor placement.



Figure 9: Proposed harness layout. The hardware illustrations simplify cable routing for visual clarity.

7 TEST & VALIDATION

7.1 Posture Detection

Independent angle reference. We will extend the cardboard hinge demonstration to a three-segment hinged fixture with rigid mounting plates and marked sensor axes. One IMU will be fixed to each plate. A digital inclinometer with specified accuracy of 0.2° or better, checked against a level surface and a mechanical protractor, will measure each plate's tilt independently. The two reference regional angles will be calculated from neighboring plate tilts using the same sign convention as the firmware.

Static tests will cover each regional angle from −30° to +30° in 10° steps, testing one hinge at a time and then

representative combined configurations. At each setting, the fixture will settle for 5 seconds and record 10 seconds of paired reference/system measurements. We will also tilt all plates together by $-20°$, $0°$, and $+20°$ to check that leaning does not create a regional bend. Both raw regional angles and baseline-subtracted deviations will be compared with their references, with bias, maximum error, and RMSE reported separately.

Dynamic (or moving) tests will use ten forward/back cycles through the same $\pm 30°$ range at approximately $10°/s$ and $30°/s$. A fixed side-view camera at 120 frames/s or faster will track straight markers attached to each plate against a leveled reference line.

Alert trials. With a test threshold of $15°$ and a 2-second persistence interval, the fixture will hold deviations of $10°$, $14°$, $16°$, and $25°$ for 1 and 4 seconds, repeated five times for each region. For the representative 20-second reminder interval, the same angles will be tested with 19-second and 25-second holds, repeated five times per region, so the trials cover both shorter and longer holds than the configured duration. Detection rate is the number of these trials producing an alert divided by all intended-alert trials, with a target of at least 90%. Motor onset will be measured with the logged trigger and an independently recorded motor-current signal or driver output, targeting at most 1 second after the persistence interval ends. Trials will also check simultaneous regional triggers, pause, sensor faults, and continued sensing during motor activity.

Haptic perceptibility. At least ten participants will each receive ten delivered 500 ms pulses at randomized intervals while wearing typical clothing. Five interleaved no-pulse intervals will check whether participants report cues that were not delivered.

7.2 Calibration Testing

Calibration and movement-check usability will be evaluated with at least ten participants across the intended fit range. Each participant will fit the harness, align the marked sensors near T3, T12, and L5, and complete five neutral-posture calibrations. We will record baseline variation, fit adjustments, completion time, and any assistance required. Participants will then choose regional and duration thresholds through the interface.

For the ordinary-movement test, each participant will perform five repetitions of six actions: talking with natural gestures, turning left/right, reaching toward the desk, reclining, standing and taking several steps, and a deliberate seated stretch. One trial starts from the calibrated position, includes one instructed action, and ends after returning to baseline. The action script and actual duration will be recorded.

The target is at least 95% without an unwanted alert under the stated configuration. The movement preview is meant as a configuration assistance rather than evidence that rejection has already been achieved.

7.3 Spine Visualization

The three-segment fixture will first verify that the display bends in the correct direction and shows the expected upper/lower regional changes. End-to-end latency will be measured with a camera at 120 frames/s or faster showing both the fixture and desktop screen. For each of 30 abrupt fixture transitions, we will count frames from the visible physical change to the first corresponding display change. The target is less than 250 ms in every tested transition. Sampling, filter lag, BLE buffering, and rendering are all included. BLE packet timestamps will help locate delays but will not replace this sensor-to-screen test. Disconnection and stale-data behavior will also be checked.

7.4 System Stress Testing and Full Day Wear Tests

The integrated system will run for at least eight hours with continuous sensing, a connected desktop at 25 updates/s, and a test motor pulse of 500 ms every 30 seconds. Current at the battery input will be logged both with the motor off and during pulses. A separate trial with normal event-driven alerts will compare realistic usage with this heavier motor test.

Total wearable mass will be measured with a scale, including the garment, battery, electronics, enclosure, and all cables.

Temperature tests will use contact probes on the user-facing electronics enclosure, battery pocket, and motor attachment at an ambient temperature of $22°C \pm 2°C$. After recording an unpowered baseline, temperatures will be logged at least once every 15 minutes during the eight-hour operating trial, including the repeated-pulse condition. Every measured contact surface must remain below $34°C$.

Wear trials will record comfort, attachment stability, helpful and unnecessary reminders, and connection failures.

8 CURRENT WORK

We have received the initial components and assembled a bench prototype connecting the Feather MCU, two LSM6DSOX IMUs, the DRV2605L driver, and one vibration motor. Figures 10 and 11 show the setup and bend demonstration. The two IMUs communicate directly at addresses 0x6A and 0x6B, so this initial test does not yet need the multiplexer. We taped the sensors to two cardboard panels joined at a hinge, bent the panels by approximately $30°$, and demonstrated that the firmware calculated a relative orientation and activated the motor.

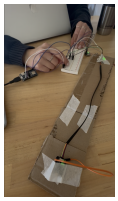


Figure 10: Two-IMU prototype connected to the Feather and haptic driver.

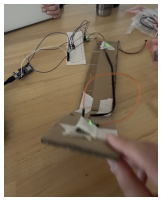


Figure 11: Prototype during forward bending.

9 PROJECT MANAGEMENT

9.1 Schedule

Figure 12 shows the project schedule, with parallel work on sensor placement, angle calculations, haptic control, and the desktop application. These tasks lead into wearable integration, accuracy and user testing, and combined software/hardware testing. The final two weeks provide us with extra buffer time for integration.

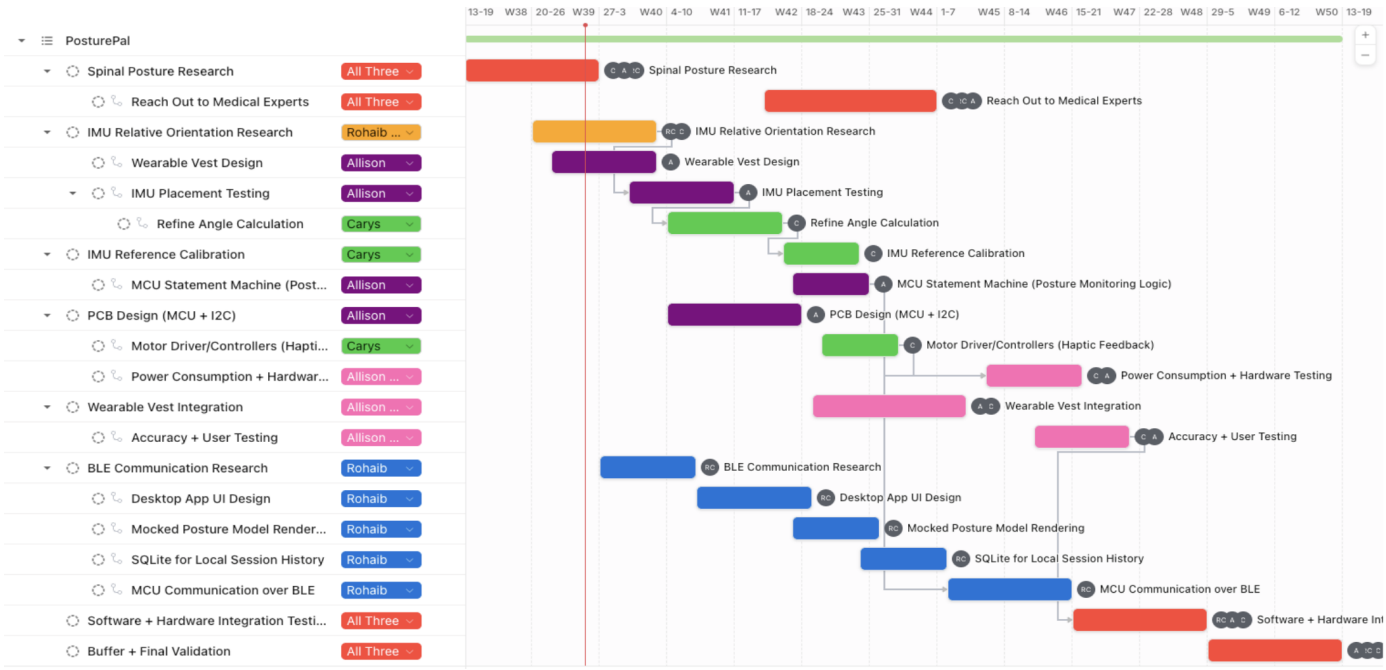


Figure 12: Team project schedule, showing task owners and dependencies.

The carrier PCB depends on reviewed power paths, connectors, and sensor addressing.

9.2 Team Member Responsibilities

Responsibilities are mostly split up by hardware and software subsystems, but all team members will be involved in spinal-posture research. Allison is working on the physical prototype: wearable vest design, IMU placement testing, MCU posture-monitoring state machine, carrier-PCB/I^2C design, and wearable integration. Carys leads angle-calculation refinement, IMU reference calibration, and haptic motor driver/controller work. Rohaib leads IMU relative-orientation research and the desktop and wireless stack, SQLite session history, and MCU-to-desktop BLE communication.

9.3 Bill of Materials and Budget

Our current bill of materials includes the IMUs, MCU, haptic driver, vibration motors, cables, and adjustable harnesses, with a recorded total of $166.63 [9]. Figure 13 shows the parts, quantities, suppliers, and costs from our team spreadsheet. We ordered extra IMUs and motors for prototyping and replacements, although the final wearable will use three IMUs and one motor.

	A	B	C	D	E	F
1	**Part**	**Description**	**Link**	**Quantity**	**Per-Item Cost**	**Total Cost**
2	LSM6DSOX	IMU	https://www.adafruit.com/product/4438	6	$11.95	$71.70
3	Stemma QT M-M	IMU->IMU Connectors	https://www.adafruit.com/product/5384	6	$1.25	$7.50
4	nRF52840	MCU	https://www.adafruit.com/product/4062	1	$24.95	$24.95
5	STEMMA QT M-F	IMU->PCB Connectors	https://www.adafruit.com/product/4397	3	$0.95	$2.85
6	DRV2605L	Haptic Motor Controller	https://www.adafruit.com/product/2305	1	$7.95	$7.95
7	Vibrating Motor Discs	Haptic Motor Controller	https://www.adafruit.com/product/1201	6	$1.95	$11.70
8	PCA9548	I2C MUX	https://www.adafruit.com/product/5626	0		
9	Battery	LiPo 3.7V 1000mAh?				
10	Vest	Mesh Vest with adjustable straps	https://www.amazon.com/Weaver-Leath	2	$19.99	$39.98
11						
12						
13						
14						
15						
16						
17						
18	**Total**					$166.63
19						

Figure 13: PosturePal bill of materials from the team's parts spreadsheet. Red columns indicate parts that aren't required for prototyping but will be on the final product (present for placeholding).

The battery and multiplexer still need to be purchased. We will also account for the carrier PCB, connectors, wiring, and materials used to secure the electronics to the vest.

9.4 TechSpark Use Plan

The team will use TechSpark sewing equipment to modify the purchased vest, create adjustable sensor mounting locations near T3, T12, and L5, secure wiring with strain relief, and integrate the electronics in a comfortable and accessible enclosure or pocket. TechSpark resources may also be used for mechanical prototyping and assembly where appropriate, with final fabrication choices dependent on available equipment and safety procedures. Other than that, work can be conducted in HH1307, where our supplies are currently being stored.

9.5 Risk Mitigation Plans

Key risks include sensor slippage, timing and orientation errors, communication failures, motor interference, and limited battery life. We will address these through secure mounts, calibration and reference-angle checks, wiring tests, and measurements of motor noise, BLE latency, and battery endurance. Haptic feedback will operate independently of the desktop connection. Each subsystem will be tested before integration, with time reserved for corrections and final validation.

10 RELATED WORK

Paloschi et al. used three magneto-inertial sensors at T3, T12, and S1 to estimate thoracic kyphosis and lumbar lordosis during forward bending and sit-to-stand tasks, validating their results against optical motion capture [3]. PosturePal adopts the neighboring-sensor comparison principle, but uses L5 for its lowest sensing location and measures personalized changes rather than claiming clinical kyphosis/lordosis measurements. Its main addition is a local sustained-deviation rule linked to haptic feedback and a desktop calibration/history interface.

Laidig and Seel's VQF method provides an established approach to orientation estimation with gyro bias estimation and acceleration-based tilt correction [7]. Its implementation supports magnetometer-free operation [8], matching the six-axis LSM6DSOX sensors selected here.

Pourhashemi et al. examined delayed visual feedback during a dynamic standing task [5]. Their results support keeping visual delay low, while Nielsen's response-time guidance provides a general interaction reference [4]. Neither source validates PosturePal's exact seated-use latency limit. PosturePal's intended interaction is a reminder that lets the user choose whether and how to adjust posture.

11 SUMMARY

PosturePal is designed to provide personalized posture awareness without mechanically forcing the user's back into a prescribed shape. Three sensing locations yield upper- and lower-back relative-angle measures, which are compared with calibrated reference values and user-selected persistence thresholds. A microcontroller performs contin-

uous detection and drives a haptic reminder, while a BLE-connected desktop application supports configuration, visualization, and history. The current two-IMU cardboard-hinge prototype demonstrates relative-orientation calculation and motor activation. The remaining work is to implement the three-sensor VQF pipeline, complete wearable and desktop integration, and experimentally verify angle accuracy, alert timing, display latency, battery endurance, and user comfort against the design requirements.

GLOSSARY OF ACRONYMS

BLE Bluetooth Low Energy

GPIO General-Purpose Input/Output

I²C Inter-Integrated Circuit

IMU Inertial Measurement Unit

LiPo Lithium-Polymer

MCU Microcontroller Unit

PCB Printed Circuit Board

RMSE Root Mean Square Error

SD Standard Deviation

SPI Serial Peripheral Interface

VQF Versatile Quaternion-based Filter

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